L1.2 Limits

Device block first, then the rest at your table. Check everything against the Quarto tonight.

3 · A general flowchart for evaluating limits — at the device

1. Open **Limit Diagnostic**. Work cards until you have logged **six**: at least **two** that return $\frac{0}{0}$, one that returns a number, one that returns $\frac{b}{0}$, and one from **below the divider**.

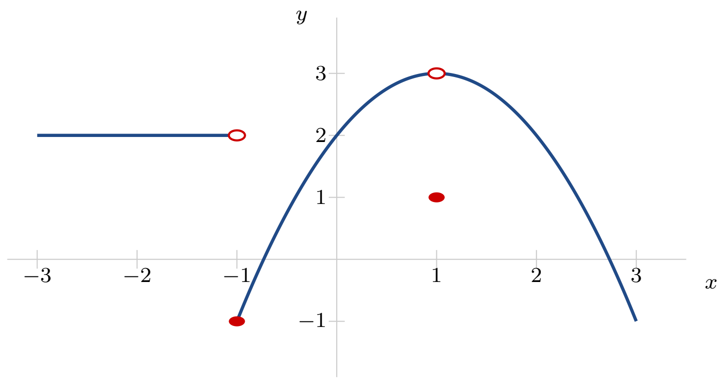
card	$f(a)$ **returned**	**which line of the chart**	**the limit**

2. One card returns a perfectly clean number and the limit still does not exist. Find it. What is true about that function near a that is not true of the other clean-number cards?

3. Below the divider, work **both** oscillation cards. They have the same zeros and crowd the same way, and only one of them has a limit. Write the one question the chart asks to separate them — and the answer for each card.

1 · What a limit is

4. From the graph:



(a) $\lim_{x \to -1^-} f(x)$ (b) $\lim_{x \to -1^+} f(x)$ (c) $\lim_{x \to -1} f(x)$ (d) $f(-1)$ (e) $\lim_{x \to 1} f(x)$ (f) $f(1)$

(g) One of the two marked x-values gets the shorthand `HE @`. Which one?

2 · Diagnosing limits

5. Run the diagnostic on each. Write what came back and what the chart says to do next — **do not evaluate yet**. Then finish the two that need no rewriting at all.

(a) $\lim_{x \to 4} \dfrac{x^2 - 3x - 4}{x - 4}$ (b) $\lim_{x \to 2} \dfrac{x^2 - 4}{x + 2}$ (c) $\lim_{x \to -2} \dfrac{x}{x + 2}$ (d) $\lim_{x \to 0} \sqrt{\dfrac{4x + 9}{x + 1}}$

6. Factor and cancel.

> **Stop.** Before you cancel — what does a cancellation leave behind on the graph, and what is the limit equal to?

(a) $\lim_{x \to 4} \dfrac{x^2 - 3x - 4}{x - 4}$ (b) $\lim_{x \to 3} \dfrac{2x^2 - 5x - 3}{x^2 - 9}$

For (b), also give the full shorthand for the graph: `HA @`, `Z:`, `HE @`, `VA @`.

7. Square-rooty, so rationalize.

$$\lim_{h \to 0} \frac{\sqrt{4 + h} - 2}{h}$$

8. Both of these return $\frac{b}{0}$. For each: LHL, then RHL, then the two-sided answer.

> **Stop.** Before you write ∞ — what does $\frac{b}{0}$ tell you, and what does it not tell you?

(a) $\lim_{x \to -2} \dfrac{x}{x + 2}$ (b) $\lim_{x \to 1} \dfrac{5}{(x - 1)^2}$

9. You are told only that

$$4x - 3 \leqslant f(x) \leqslant x^2 + 1 \qquad \text{for every } x.$$

Find $\lim_{x \to 2} f(x)$, and say why you never needed a formula for f.

4 · The laws

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10. Let

$$f(x) = 3x - 1 \quad (x \neq 1) \qquad\qquad g(x) = \begin{cases} (x-1)^2 & x \neq 2 \\ 0 & x = 2 \end{cases}$$

Evaluate:

(a) $\lim_{x \to 2} \left[4f(x) - g(x)\right]$ (b) $\lim_{x \to 1} \sqrt{3f(x)}$ (c) $\lim_{x \to 0} \dfrac{f(x)}{g(x)}$

11. Same f and g: $\lim_{x \to 1} \dfrac{f(x)}{g(x)}$. The quotient law does **not** apply here — which hypothesis fails? Then go back to the function itself and work both sides.